Letter of Motivation

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To whom it may concern,

I have been graduated from **Department of Civil Engineering Middle East Technical University (METU)**, Ankara Turkey, at 1999. I was very enthusiastic and successful about courses on mechanics and numerical methods. During the third-year undergraduate course **CE384-Structural Analysis**, I have been first introduced to finite element method. As part of the course we have developed a finite element program for the analysis of 2D frame structures using Mathcad. After I have graduated, I have been employed as a teaching assistant at Structural Mechanics Division of the same department for about 2 years. I was also accepted to the same division to pursue my master study. During graduate and undergraduate studies at METU, although not compulsory, I have taken overall 5 courses at master level from Department of Engineering Science offering courses mainly on theoretical and computational mechanics. At 2002, I have been awarded MS degree from Department of Civil Engineering METU under the supervision of Prof. Uğur Polat and Prof. Yalçın Mengi. The title of my thesis was **"Interaction Analysis for the response of Structures Subjected to Seismic Exications: A Parametric Study"**.

At 2001, I have been accepted to **COMMAS (Computational Mechanics of Materials and Structures)** master program organized by Stuttgart University. I was ranked first in my class with a grade point average of 1.0. During the program I have been introduced to Matlab. I have developed my own finite element program for 1D and 2D continua using Matlab. I have further improved this program later on and used it extensively in my research. In time, it has become an interactive command based program together with some useful visualization capabilities. I have done my master thesis at the **Institute of Applied Mechanics (Chair-I)** under the supervision of Prof. Christian Miehe and Dr. Ercan Gürses [1]. The title of my thesis was **"Comparison of alternative kinematical approaches to finite element formulations with embedded strong discontinuities"**. In this study I have implemented 2D finite element formulations for elements cut by strong discontinuities, like cracks, based on elemental and nodal enrichments. Thermodynamically consistent framework for the cohesive surfaces of displacement discontinuities has been developed based on energy equation only,

First Law, without resorting to entropy inequality, Second Law. I have developed a really original rederivation by myself to motivate locking free finite element formulation, namely statically and kinematically optimal nonsymmetric (SKON) formulation, based on geometrical interpretation of spatial derivate of nodal shape functions. I have also implemented a damage type traction-separation constitutive law for tractions transmitted through cohesive macroscopic cracks based on internal variable formulation and data structure and algorithms for the global tracking of evolving strong discontinuities. My implementations have been verified by repeating simulations presented in [2]. Finally note that all of the above mentioned implementations have been successfully realized in CMP, an extension of FEAP program, developed and maintained at the institute.

After I graduated from COMMAS program, I have been offered a research assistantship position by Prof. Wolfgang A. Wall at the **Institute for Computational Mechanics Technical University of Munich** to pursue my PhD study. There I have implemented coupled Levelset-XFEM formulation for the simulation of 2D two-phase fluid flows. In addition, I have conducted the diploma thesis of Dr. Peter Gamnitzer, in collaboration with Prof. Wolfgang A. Wall, on overlapping domain decomposition methods for flow simulations [3,4]. All of the above mentioned formulations have been successfully implemented into finite element program CARAT developed and maintained at the institute, at that time together with **Institute for Structural Mechanics University of Stuttgart** directed by Prof. Ekkehard Ramm. The program was written in C programming language.

At 2005, I have joined to **Institute of Construction Materials University of Stuttgart** to pursue my PhD study. There I have worked on finite element formulations for contact and mesh adaptivity. I have developed a finite element program for the implicit simulation of large-deformation frictionless contact based on Penalty and Augmented-Lagrangian formulations together with adaptive-mesh-refinement and adaptive-remeshing algorithms in 2D and 3D using Matlab. Then I have transferred 3D part of this program together with contact algorithms and remeshing into Fortran [5-8]. Note that 3D node-to-surface finite elements for frictionless contact have been developed based on derivations presented in [9]. Also, for pre- and post- processing tasks and to perform remeshing, I have developed an interface to Femap program by using api interface provided based on Femap BASIC scripting language.

From 2008 to 2018, I have worked at a software company, **PROYA** Ankara Turkey, as a software engineer. I was the member of a software team gathered to develop COMOSYS (Construction Modelling Systems) program, an object-oriented software for the analysis, design and detailing of commercial and industrial buildings. In COMOSYS, I was responsible for 3D rebar modelling

module [10,11] which has been successfully used in several projects [12]. The whole structure of the module has been constructed by me and I was the only person working on it until I left. For this purpose, I have developed GUI, topological data structures and algorithms for 3D positioning of reinforcement bars. At PROYA, I have learned object-oriented programming using C++ programming language, GUI development, basic OpenGL programming, using STL containers and employing Winapi functions to interact with windows operating system.

At 2011, I have been enrolled as a PhD student at the **Institute of Construction Materials University of Stuttgart** under the supervision of Prof. Josko Ozbolt. After successfully defending my thesis I have been awarded Dr.-Ing. degree at 2014 [13]. During the period of my study I was in Turkey. I have come to Stuttgart periodically to discuss the current progress and future work. The title of my thesis was **"High velocity impact and fragmentation of concrete: numerical simulation"**. Our simulations on perforation of circular plain-concrete slabs with a projectile was the only one in the literature which was able to show perforation limit documented in the experiments [14]. Within the context of high velocity contact-impact simulations, main ingredients of the finite element program are

- updated Lagrangian finite element formulation,
- explicit time integration,
- predictor-corrector type explicit/implicit contact algorithm,
- element deletion(erosion) algorithm,
- rate and temperature dependent microplane material model at large strains.

During my PhD study, I have developed a finite element analysis program, CIF (Contact Impact and Fragmentation), for 3D high velocity contact impact events [15]. For this purpose, I have developed a program in the form of DLL to perform finite element computations using Intel Fortran. In the program, topological data structures are constructed from underlying finite element discretizations and used effectively in contact algorithms on eroding surfaces. For pre and post processing tasks I have developed a 32 bit windows program, a GUI, using Borland C++ by employing built-in VCL, visual component library. In GUI, rendering and selection algorithms for finite element entities, namely nodes and elements, have been implemented using OpenGL. In CIF, post processing options in the form of time history plot, deformed shape and combination of both are available. Also, finite element computations are performed on a separate thread so that interactive monitoring of the results, while the simulation is running, is possible. The last step in the development of CIF was to take the material models out and put them into a material library in the form of DLLs. With this final enhancement it would be possible to add a new material model and use it with CIF. I have also developed another program, CIF-JOB, basically to organize simulation projects and perform

parametric studies using Borland C++ [16]. In CIF-JOB, use of Winapi functions is made to copy, clean, browse, run and terminate simulation projects. Simulation projects could be run in either silent mode (without GUI) or interactive mode (with GUI) and it is possible to perform interactive monitoring of multiple simulation projects on the same platform, which removes all intermediate repetitive steps to visualize results from different simulations together rendering an error-free, extremely fast user-friendly mechanism. Finally note that CIF and CIF-JOB have been used successfully in several occasions by the members of the institute [17-19].

After I have obtained my PhD degree, I have given three undergraduate courses, 2 courses on Strength of Materials and one course on Numerical Methods, as part-time instructor at different universities in Ankara Turkey.

During the last year, 08.2018-07.2019, I have worked at the **Institute of Construction Materials University of Stuttgart** as a research associate. It was a good opportunity for me to combine my past experiences from computational mechanics and programming in an optimal way. During this period, I have developed a really general purpose finite element program for the simulation of different physical phenomena [20-22]. **Note that underlying framework is more general and can be used to develop a software for a completely different purpose.** Until I left, I have implemented simulation modules for heat transfer, solid mechanics, explicit dynamics and coupled thermo-mechanics. The program is highly modular and all modules,

- simulations,
- element formulations,
- material models,
- special algorithms (plugins), like contact, element erosion, explicit time integration,
- finite element utilities, like boundary conditions, time curves, initial conditions,

have been implemented in the form of DLLs. The interaction between these different modules has been achieved by client-server based message passing interface, which shares very common features with low level messaging mechanism used in windows operating system. This modular approach enables the continuous independent development of the program in different disciplines, like new simulations, element formulations, material models etc., within a team of programmers (researchers). The underlying structure also enjoys another very useful feature so called code safety (hiding). Finally, please note that the above-mentioned structure has been constructed for both processor developed in Intel Fortran and pre-post processor developed in Borland C++.

The main driving force (constraint) while reaching to this important milestone mentioned above was to end up with a flexible mechanism which serves as a general framework for developers to realize (test) their new ideas or to integrate their own past developments easily with little possible effort. It is also noteworthy to point out that in the Fortran side, some modern features like derived types (objects) with member functions, almost everywhere, implementations based on object-oriented programming principles, at some very limited places for now, have been utilized. Heavy use of data and function pointers and functions (subroutines) like TRANSFER, LOC, C_F_PROCPOINTER has been made to end up with this highly modular flexible C-like structure. Of course, in achieving this my past experiences about C++ programming language helped me quite a lot.

To conclude, in which road I want to proceed is very much evident from this brief overview. I have a very sound understanding of computational mechanics from both theoretical and practical (implementation) point of views. In the past, I have been always in touch with either computational mechanics or programming. I have developed programs using Matlab, Fortran and C++ [7,15,16,20-23]. I was initiative and self-starter in all the programs that I have developed. I am patient. I have a dedicated and goal-oriented attitude. Based on my past experiences, as a student and as a professional, I am very competitive in all aspects related with computational mechanics and programming.

Sincerely yours,

Barış İrhan.

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